



# Yakeen Plus (2025)

## SHORT PRACTICE TEST - 06

DURATION: 60 Minutes

DATE: 20/10/2024

M.MARKS : 192

### Topics Covered

|                   |                                                                                                                                                                                                                         |
|-------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Physics:</b>   | Oscillations, Waves, Electric Charges and Fields, Electrostatic Potential and Capacitance                                                                                                                               |
| <b>Chemistry:</b> | Organic Chemistry: Some Basic principles and Techniques (IUPAC Naming), Organic Chemistry: Some Basic principles and Techniques (Isomerism), Organic Chemistry: Some Basic principles and Techniques (GOC), Hydrocarbon |
| <b>Biology:</b>   | <b>(Botany)</b> : Anatomy of Flowering Plants, Respiration in Plants<br><b>(Zoology)</b> : Neural Control & Coordination, Chemical Coordination & Integration                                                           |

#### General Instructions:

1. Immediately fill in the particulars on this page of the test booklet.
2. The test is of **60 minutes** duration.
3. The test booklet consists of **48** questions. The maximum marks are **192**.
4. All questions are compulsory.
5. There is only one correct response for each question.
6. Each correct answer will give **4** marks while **1** Mark will be deducted for a wrong MCQ response.
7. No student is allowed to carry any textual material, printed or written, bits of papers, pager, mobile phone, any electronic device, etc. inside the examination room/hall.
8. On completion of the test, the candidate must hand over the Answer Sheet to the Invigilator on duty in the Room/Hall. However, the candidates are allowed to take away this Test Booklet with them.
9. **Do not fold or make any stray mark on the Answer Sheet (OMR).**

#### OMR Instructions:

1. Use blue/black dark ballpoint pens.
2. Darken the bubbles completely. Don't put a tick mark or a cross mark where it is specified that you fill the bubbles completely. Half-filled or over-filled bubbles will not be read by the software.
3. Never use pencils to mark your answers.
4. Never use whiteners to rectify filling errors as they may disrupt the scanning and evaluation process.
5. Writing on the OMR Sheet is permitted on the specified area only and even small marks other than the specified area may create problems during the evaluation.
6. Multiple markings will be treated as invalid responses.
7. **Do not fold or make any stray mark on the Answer Sheet (OMR).**

Name of the Student (In CAPITALS) : \_\_\_\_\_

Roll Number : \_\_\_\_\_

OMR Bar Code Number : \_\_\_\_\_

Candidate's Signature : \_\_\_\_\_ Invigilator's Signature \_\_\_\_\_

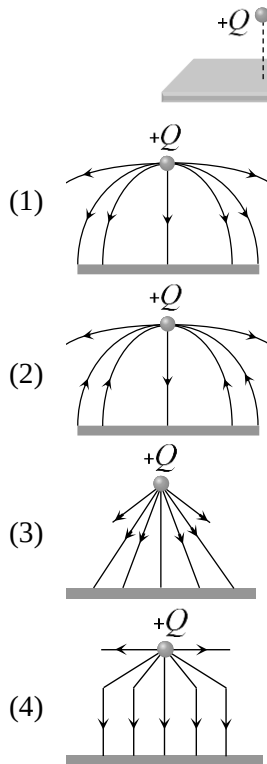
## SECTION-(I) PHYSICS

1. The phase difference between two points separated by 1m in a wave of frequency 120 Hz is  $90^\circ$ . The wave velocity is:  
(1) 180 m/s (2) 240 m/s  
(3) 480 m/s (4) 720 m/s

2. An arc of radius  $r$  carries charge. The linear density of charge is  $\lambda$  and the arc subtends a angle  $\frac{\pi}{3}$  at the centre. What is electric potential at the centre?

- (1)  $\frac{\lambda}{4\epsilon_0}$  (2)  $\frac{\lambda}{8\epsilon_0}$   
(3)  $\frac{\lambda}{12\epsilon_0}$  (4)  $\frac{\lambda}{16\epsilon_0}$

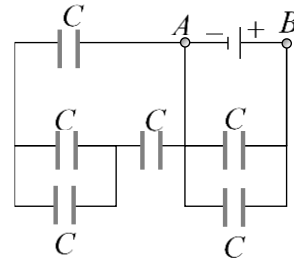
3. A charge  $Q$  is fixed at a distance  $d$  in front of an infinite metal plate. The lines of force are represented by:



4. The temperature at which the speed of sound in air becomes double of its value at  $0^\circ\text{C}$  is:  
(1) 273K (2) 546K  
(3) 1092K (4) 0K

5. A particle executing simple harmonic motion along  $y$ -axis has its motion described by the equation  $y = A\sin(\omega t) + B$ . The amplitude of the simple harmonic motion is:  
(1)  $A$   
(2)  $B$   
(3)  $A + B$   
(4)  $\sqrt{A+B}$

6. Find equivalent capacitance between A and B.



- (1)  $6C$  (2)  $5C$   
(3)  $3C$  (4)  $2C$

7. The radii of two metallic spheres  $P$  and  $Q$  are  $r_1$  and  $r_2$  respectively. They are given the same charge. If  $r_1 > r_2$ , then on connecting them with a thin wire, the charge will flow.

- (1) from  $P$  to  $Q$ .  
(2) from  $Q$  to  $P$ .  
(3) neither the charge will flow from  $P$  to  $Q$  nor from  $Q$  to  $P$ .  
(4) the information is incomplete.

8. The equation of a wave is  $y = 2\sin\pi(0.5x - 200t)$ , where  $x$  and  $y$  are expressed in cm and  $t$  in sec. The wave velocity is:

- (1) 100 cm/sec  
(2) 200 cm/sec  
(3) 300 cm/sec  
(4) 400 cm/sec

9. If there are  $n$  capacitors in parallel connected to  $V$  volt source, then the energy stored is equal to:

- (1)  $CV$  (2)  $\frac{1}{2}nCV^2$   
(3)  $CV^2$  (4)  $\frac{1}{2n}CV^2$

10. An infinite non-conducting sheet has a surface charge density  $\sigma = 0.10 \mu\text{C}/\text{m}^2$  on one side. How far apart are equipotential surfaces whose potentials differ by 50 V.

- (1) 8.85 m (2) 8.85 cm  
(3) 8.85 mm (4) 88.5 mm

11. The velocity of a particle in simple harmonic motion at displacement  $y$  from mean position is:

- (1)  $\omega\sqrt{a^2 + y^2}$  (2)  $\omega\sqrt{a^2 - y^2}$   
(3)  $\omega y$  (4)  $\omega^2\sqrt{a^2 - y^2}$

12. A particle executing simple harmonic motion with amplitude of 0.1 m. At a certain instant when its displacement is 0.02 m, its acceleration is  $0.5 \text{ m/s}^2$ . The maximum velocity of the particle is: (in m/s)

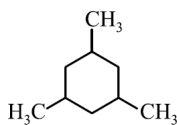
- (1) 0.01 (2) 0.05  
(3) 0.5 (4) 0.25

## SECTION-(II) CHEMISTRY

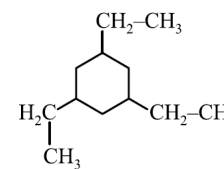
13. Which isomer of hexane has only two different sets of structurally equivalent hydrogen atoms?

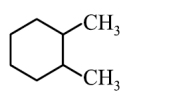
- (1) 2, 2-dimethylbutane
- (2) 2-methylpentane
- (3) 3-methylpentane
- (4) 2, 3-dimethylbutane

14. In which compound  
1°C: 2°C: 3°C (carbon) = 1:1:1?

- 

(1)



(2)
- 

(3)



(4)

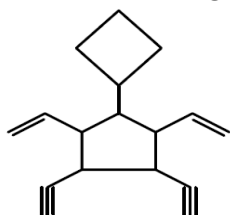
15. The strongest acid among the following compounds is:

- (1)  $\text{HOOC}-(\text{CH}_2)_2-\text{COOH}$
- (2)  $\text{H}_3\text{N}^+-(\text{CH}_2)_2-\text{COOH}$
- (3)  $\text{F}-(\text{CH}_2)_2-\text{COOH}$
- (4)  $\text{H}_3\text{CO}-(\text{CH}_2)_2-\text{COOH}$

16. The hyperconjugative stabilities of tert-butyl cation and 2-butene, respectively, are due to:

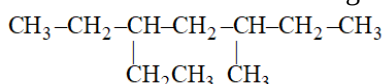
- (1)  $\sigma \rightarrow p$  (empty) and  $\sigma \rightarrow \pi^*$  electron delocalizations
- (2)  $\sigma \rightarrow \sigma^*$  and  $\sigma \rightarrow \pi$  electron delocalizations
- (3)  $\sigma \rightarrow p$  (filled) and  $\sigma \rightarrow \pi$  electron delocalizations
- (4)  $p$  (filled)  $\rightarrow \sigma^*$  and  $\sigma \rightarrow \pi^*$  electron delocalizations

17. Degree of unsaturation in the given compound is:



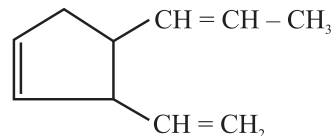
- (1) 8
- (2) 16
- (3) 10
- (4) 12

18. Correct IUPAC name of the given compound is:



- (1) 5-Ethyl-3-methylheptane
- (2) 1,1-Diethyl-3-methylpentane
- (3) 3-Methyl-5-ethylheptane
- (4) 3-Ethyl-5-methylheptane

19. Total number of stereoisomers of the given compound is:



- (1) 6
- (2) 8
- (3) 16
- (4) 4

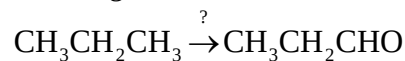
20. Which of the following is an electrophilic reagent?

- (1)  $\text{RO}^-$
- (2)  $\text{BF}_3$
- (3)  $\text{NH}_3$
- (4)  $\text{ROH}$

21. Sodalime decarboxylation of sodium propionate produces

- (1) Propane
- (2) Ethane
- (3) Methane
- (4) Butane

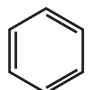
22. Which of the following reagent is used for the following conversion?



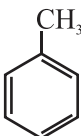
- (1) Manganese acetate
- (2) Molybdenum oxide
- (3) Potassium permanganate
- (4) Copper at high temperature and pressure

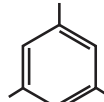
23.  $3\text{CH}_3-\text{C} \equiv \text{CH} \xrightarrow[\text{Cu tube}]{\text{Red Hot}} \text{P}$ ,

P is:

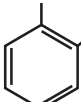
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(1)



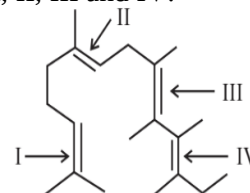
(2)
- 

(3)



(4)

24. In the following structure, the double bonds are marked as I, II, III and IV:



Geometrical isomerism is **not** possible at which site(s)?

- (1) I
- (2) III
- (3) I and III
- (4) III and IV

## SECTION-(III) BOTANY

25. Select the **incorrect** match from the following pair.

- (1) Radial vascular bundle : Dicotyledonous root
- (2) Polyarch xylem : Dicotyledonous root
- (3) Small inconspicuous pith : Dicotyledonous root
- (4) Well developed pith : Monocotyledonous root

26. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:

**Assertion A:** In both lactic acid and alcohol fermentation less than seven per cent of the energy in glucose is released.

**Reason R:** Net gain of ATP in fermentation is 8 ATP.

In the light of the above statements, choose the **correct** answer from the options given below:

- (1) A is true but R is false.
- (2) A is false but R is true.
- (3) Both A and R are true and R is the correct explanation of A.
- (4) Both A and R are true but R is NOT the correct explanation of A.

27. Which one is **correct** about the products formed by single turn of citric acid cycle?

- |                      |                        |
|----------------------|------------------------|
| A. 2 NADH            | B. 1 FADH <sub>2</sub> |
| C. 3 CO <sub>2</sub> | D. 1 GTP               |

Choose the most appropriate answer from the options given below:

- (1) A and B only
- (2) B and D only
- (3) A, B and C only
- (4) A, C and D only

28. Read the following statements and select the **correct** option for **A**, **B** and **C** respectively.

(I)     **[A]**     consists of long, narrow cells with thick and lignified cell walls having a few or numerous pits.

(II)     **[B]**     consists of cells which are much thickened at the corners due to a deposition of cellulose, hemicellulose and pectin.

(III) Thick-walled, elongated and pointed cells, generally occurring in groups, in various parts of the plant are     **[C]**    .

- | <b>[A]</b>       | <b>[B]</b>   | <b>[C]</b>  |
|------------------|--------------|-------------|
| (1) Collenchyma  | Sclerenchyma | fibres      |
| (2) Sclerenchyma | Collenchyma  | fibres      |
| (3) Sclerenchyma | Fibres       | collenchyma |
| (4) Vessels      | Fibres       | collenchyma |

29. Select the **incorrect** statement.

- (1) Cytochrome c is a small protein attached to the outer surface of the inner membrane of mitochondria.
- (2) Complex IV refers to cytochrome c oxidase complex.
- (3) Ubiquinone receives reducing equivalents via FADH<sub>2</sub> only.
- (4) Production of ATP from ADP and inorganic phosphate occurs through complex V.

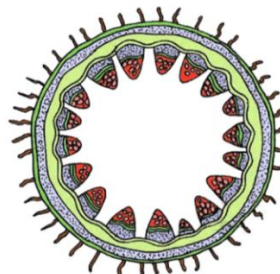
30. Casparian strip is made up of:

- |                |              |
|----------------|--------------|
| (1) lignin.    | (2) suberin. |
| (3) cellulose. | (4) Pectin.  |

31. Identify the site of glycolytic pathway in case of organism undergoing aerobic and anaerobic respiration.

- (1) Mitochondria in both anaerobes and aerobes.
- (2) Mitochondria in anaerobes whereas cytoplasm in aerobes.
- (3) Cytoplasm in anaerobes whereas mitochondria in aerobes.
- (4) Cytoplasm in both anaerobes and aerobes.

32. The following diagram is a transverse section of:



- (1) Dicotyledonous root.
- (2) Dicotyledonous stem.
- (3) Monocotyledonous root.
- (4) Monocotyledonous stem.

33. Given below are two statements.

**Statement I:** Glucose is the favoured substrate for respiration.

**Statement II:** It would be better to consider the respiratory pathway as a catabolic one rather than as an amphibolic pathway.

In the light of the above statements, choose the most appropriate answer from the options given below:

- (1) Statement I is correct but Statement II is incorrect.
- (2) Statement I is incorrect but Statement II is correct.
- (3) Both Statement I and Statement II are correct.
- (4) Both Statement I and Statement II are incorrect.

34. Select the **correct** match w.r.t. respiratory quotient (RQ).
- (1) Glucose  $\rightarrow$  1
  - (2) Protein  $\rightarrow$  0.7
  - (3) Tripalmitin  $\rightarrow$  0.9
  - (4) Fats  $\rightarrow$  More than 1
35. Read the statements carefully and choose the **correct** option.
- A. In grasses, certain adaxial epidermal cells along the veins modify themselves into large and empty cells. These are called bulliform cells.
  - B. Bulliform cells in the monocot leaves make the leaf surface exposed.
  - C. Bulliform cells in the monocot leaves make the leaves curl inwards.

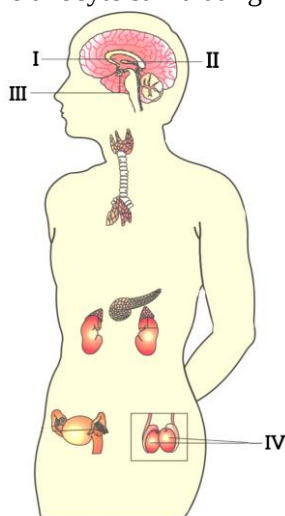
- D. Isobilateral leaf shows parallel venation. Choose the *most appropriate* answer from the options given below:
- (1) A, B and C only
  - (2) A and C
  - (3) B only
  - (4) A, B, C and D.

36. Stomatal apparatus consists of:
- (1) subsidiary cells.
  - (2) guard cells.
  - (3) stomatal aperture.
  - (4) All of these

[Short Practice Test-06 | Yakeen Plus (2025) | 20/10/2024]  
Neural Control & Coordination Chemical Coordination & Integration

## SECTION-(IV) ZOOLOGY

37. Which of the following statements is **correct** regarding unmyelinated nerve fibers?
- (1) They are enveloped by multiple Schwann cells.
  - (2) They are commonly found only in the central nervous system.
  - (3) They are enclosed by a Schwann cell but lack a myelin sheath.
  - (4) They are enveloped by a single Schwann cell.
38. Refer to the given diagram and choose the **correct** option for the gland which is responsible for releasing melanocyte stimulating hormone?



- (1) I
  - (2) II
  - (3) III
  - (4) IV
39. What happens immediately after the influx of  $\text{Na}^+$  ions during the generation of an action potential?

- (1) The outer surface of the membrane becomes positively charged.
- (2) The inner surface of the membrane becomes negatively charged.
- (3) An efflux of potassium ions occurs.
- (4) The membrane at the stimulated site becomes depolarised.

40. Read the following statements (A-E).
- A. Hypothalamic hormones reach the pituitary gland through a portal circulatory system and regulate the functions of the anterior pituitary.
  - B. The pineal gland is located in a bony cavity called sella tursica.
  - C. Impairment affecting synthesis of antidiuretic hormone (ADH), might result in diabetes mellitus.
  - D. Thyrocalcitonin (TCT), regulates the blood calcium levels.
  - E. The process of bone resorption (dissolution/demineralisation) is stimulated by parathyroid hormone.

How many of the above statements are **incorrect**?

- (1) Two
  - (2) Three
  - (3) Four
  - (4) Five
41. All the given statements are true about cortisol, **except**,
- (1) cortisol stimulates the RBC production.
  - (2) cortisol is the main mineralocorticoid in our body.
  - (3) cortisol produces anti – inflammatory reactions.
  - (4) cortisol is involved in carbohydrate metabolism.

42. Given below are two statements:

**Statement I:** In the inner part of the cerebral cortex, the fibres of the tracts are covered with myelin sheath and are called grey matter.

**Statement II:** The association areas in the cerebral cortex are responsible for complex functions like memory and communication.

In the light of the above statements, choose the *most appropriate* answer from the options given below:

- (1) Statement I is correct but Statement II is incorrect.
- (2) Statement I is incorrect but Statement II is correct.
- (3) Both Statement I and Statement II are correct.
- (4) Both Statement I and Statement II are incorrect.

43. Match **List-I** with **List-II**.

| List-I |                       | List-II |                    |
|--------|-----------------------|---------|--------------------|
| (A)    | Amino acid derivative | (I)     | Tetraiodothyronine |
| (B)    | Peptide hormone       | (II)    | Epinephrine        |
| (C)    | Steroid hormone       | (III)   | Glucagon           |
| (D)    | Iodothyronine         | (IV)    | Progesterone       |

Choose the **correct** answer from the options given below:

- (1) A-III, B-II, C-I, D-IV
- (2) A-I, B-III, C-II, D-IV
- (3) A-II, B-III, C-IV, D-I
- (4) A-III, B-I, C-II, D-IV

44. What is the primary role of the hypothalamus in the brain?

- (1) To process visual information.
- (2) To serve as a major coordinating center for sensory information.
- (3) To regulate autonomic functions such as thirst and hunger.
- (4) Both (1) and (2)

45. Which of the following structures of the brain forms the brainstem?

- (1) Medulla oblongata, cerebellum and midbrain.
- (2) Medulla oblongata, pons and midbrain.
- (3) Midbrain, thalamus and pons.
- (4) Cerebellum, pons and medulla oblongata.

46. Find the **incorrect** match w.r.t the glands and their secretions.

- (1) Pituitary gland – Estrogen
- (2) Hypothalamus – Somatostatin
- (3) Pineal gland – Melatonin
- (4) Adrenal gland – Catecholamines

47. What is the primary advantage of impulse transmission at electrical synapses compared to chemical synapses?

- (1) Electrical synapses can amplify signals.
- (2) Electrical synapses always rely on neurotransmitter release to transmit impulses.
- (3) Electrical synapses are slower than chemical synapses in transmitting impulses.
- (4) Electrical synapses can allow direct electrical current flow between neurons.

48. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:

**Assertion A:** Glucagon is a hyperglycemic hormone.

**Reason R:** Glucagon reduces the cellular glucose uptake and utilisation.

In the light of the above statements, choose the **correct** answer from the options given below:

- (1) A is true but R is false.
- (2) A is false but R is true.
- (3) Both A and R are true and R is the correct explanation of A.
- (4) Both A and R are true but R is not the correct explanation of A.

